# Research Gap Analysis: Improving InstaSHAP via Three Targeted Innovations

Phase 3 — Covertype Dataset Study

IDENTIFIED GAPS IN INSTASHAP (ICLR 2025)

Gap 1 [CRITICAL]: Zero-Masking Off-Distribution
- x*mask creates invalid categorical states
- Zeroed one-hot groups = impossible inputs
- Surrogate learns wrong value function

Gap 2 [HIGH]: Uniform Coalition Difficulty
- All mask sizes treated equally from epoch 1
- No progressive complexity → slow convergence

Gap 3 [HIGH]: Single-Surrogate Fragility
- One surrogate cascades all errors
- No confidence signal on explanations

PROPOSED INNOVATIONS

Innovation 1: Empirical-Background Masking
- Replace absent features with real training rows
- Preserves one-hot validity + marginal distribution
- Ref: Aas et al. 2019, ViaSHAP ICML 2025

Innovation 2: Curriculum-Weighted Training
- 3-phase schedule: warm-up → standard → hard
- Temperature-controlled Shapley kernel
- Ref: Bengio 2009, ICLR 2026 Coalition Sampling

Innovation 3: Multi-Surrogate Ensemble
- Train 3 surrogates, average predictions
- Variance → explanation confidence score
- Ref: Explanation Multiplicity 2026

**Results Summary (mean ± std across seeds)**

| Variant | Accuracy | Expl MSE | Spearman ρ |
|---|---|---|---|
| zero | 0.4775±0.0000 | 0.087649 | 0.3038 |
| bg | 0.5025±0.0000 | 0.128721 | 0.0924 |
| curriculum | 0.4875±0.0000 | 0.126864 | 0.3027 |
| **full** | **0.5088±0.0000** | **0.110830** | **0.3716** |